

PRIMITIVE SETS ABOVE x

GPT-5.4 PRO

ABSTRACT. We prove that if $A \subset \mathbb{N} \cap [x, \infty)$ is primitive, then $\sum_{a \in A} (a \log a)^{-1} \leq 1 + O(1/\log x)$ as $x \rightarrow \infty$. In particular, every primitive set supported on $[x, \infty)$ satisfies the bound $1 + o(1)$. The argument constructs a sub-Markov chain on the divisibility poset whose visiting probabilities are exactly proportional to $1/(n \log n)$. The proof is elementary and self-contained.

1. INTRODUCTION

A set $A \subset \mathbb{N}$ is called *primitive* if no two distinct elements of A divide one another. Define

$$S(x) := \sup \left\{ \sum_{a \in A} \frac{1}{a \log a} : A \subset \mathbb{N} \cap [x, \infty) \text{ primitive} \right\},$$

where all logarithms are natural. The purpose of this note is to determine the correct limiting upper bound for $S(x)$.

Theorem 1. *As $x \rightarrow \infty$,*

$$S(x) \leq 1 + O\left(\frac{1}{\log x}\right).$$

Equivalently, if $A \subset \mathbb{N} \cap [x, \infty)$ is primitive, then

$$\sum_{a \in A} \frac{1}{a \log a} < 1 + o(1).$$

Since $\mathbb{N} \cap [x, \infty) = \mathbb{N} \cap [[x], \infty)$, we may replace x by $[x]$ whenever convenient.

2. PROOF OF THE THEOREM

Let Λ denote the von Mangoldt function; thus $\Lambda(q) = \log p$ if $q = p^k$ is a prime power, and $\Lambda(q) = 0$ otherwise. We shall repeatedly use the elementary identity

$$(1) \quad \sum_{q|n} \Lambda(q) = \log n \quad (n \in \mathbb{N}),$$

where the sum runs over all positive divisors of n .

The first two lemmas provide an elementary proof of the classical estimate $\sum_{q \leq t} \Lambda(q)/q = \log t + O(1)$.

Lemma 2. *One has*

$$\psi(x) := \sum_{n \leq x} \Lambda(n) \ll x \quad (x \geq 2).$$

Proof. Let

$$\vartheta(x) := \sum_{p \leq x} \log p.$$

If $n \geq 1$, every prime $p \in (n, 2n]$ divides $\binom{2n}{n}$, so

$$\vartheta(2n) - \vartheta(n) \leq \log \binom{2n}{n} \leq 2n \log 2.$$

Fix $x \geq 2$, and choose k so that $2^{k-1} < x \leq 2^k$. Then

$$\vartheta(x) \leq \vartheta(2^k) = \sum_{j=0}^{k-1} (\vartheta(2^{j+1}) - \vartheta(2^j)) \leq \sum_{j=0}^{k-1} 2^{j+1} \log 2 < 2^{k+1} \log 2 \leq 4x \log 2.$$

Therefore

$$\psi(x) = \sum_{r \geq 1} \vartheta(x^{1/r}) \leq 4 \log 2 \sum_{\lfloor \log_2 x \rfloor \geq r \geq 1} x^{1/r}.$$

The term $r = 1$ is $4x \log 2$, and

$$\sum_{\lfloor \log_2 x \rfloor \geq r \geq 2} x^{1/r} \leq (\log_2 x) x^{1/2} \ll x,$$

so $\psi(x) \ll x$. □

Lemma 3. For $t \geq 2$,

$$A(t) := \sum_{q \leq t} \frac{\Lambda(q)}{q} = \log t + O(1).$$

Proof. For an integer $N \geq 1$,

$$\log N! = \sum_{m \leq N} \Lambda(m) \left\lfloor \frac{N}{m} \right\rfloor.$$

Indeed, both sides equal $\sum_{p^k \leq N} \lfloor N/p^k \rfloor \log p$. Dividing by N yields

$$\sum_{m \leq N} \frac{\Lambda(m)}{m} = \frac{\log N!}{N} + \frac{1}{N} \sum_{m \leq N} \Lambda(m) \left\{ \frac{N}{m} \right\},$$

where $\{\cdot\}$ denotes the fractional part. By Lemma 2,

$$0 \leq \frac{1}{N} \sum_{m \leq N} \Lambda(m) \left\{ \frac{N}{m} \right\} \leq \frac{\psi(N)}{N} \ll 1.$$

Also,

$$\int_1^N \log t \, dt \leq \log N! \leq \log N + \int_1^N \log t \, dt,$$

so

$$\frac{\log N!}{N} = \log N - 1 + O\left(\frac{\log N}{N}\right) = \log N + O(1).$$

Hence

$$\sum_{m \leq N} \frac{\Lambda(m)}{m} = \log N + O(1).$$

By monotonicity, the same estimate holds for real $t \geq 2$. □

Lemma 4. Uniformly for $m \geq 1$ and $y \geq 2$ with $my \rightarrow \infty$,

$$T(m, y) := \sum_{q \geq y} \frac{\Lambda(q)}{q \log^2(mq)} = \frac{1}{\log(my)} + O\left(\frac{1}{\log^2(my)}\right).$$

Proof. Let

$$g(t) := \frac{1}{\log^2(mt)} \quad (t \geq y).$$

For $T > y$, partial summation gives

$$\sum_{y \leq q \leq T} \frac{\Lambda(q)}{q} g(q) = A(T)g(T) - A(y^-)g(y) - \int_y^T A(t)g'(t) \, dt,$$

where $A(y^-) = \lim_{u \uparrow y} A(u)$. By Lemma 3, $A(t) = \log t + O(1)$, and since

$$g'(t) = -\frac{2}{t \log^3(mt)},$$

we may let $T \rightarrow \infty$ and obtain

$$T(m, y) = -(\log y)g(y) - \int_y^\infty (\log t)g'(t) \, dt + O\left(g(y) + \int_y^\infty |g'(t)| \, dt\right).$$

The main term equals

$$\int_y^\infty \frac{dt}{t \log^2(mt)} = \frac{1}{\log(my)}.$$

Moreover,

$$g(y) + \int_y^\infty |g'(t)| dt \ll \frac{1}{\log^2(my)}.$$

This proves the claim. $\square$

Proof of Theorem 1. It is enough to prove the stated estimate for all sufficiently large x . Replacing x by $\lceil x \rceil$, we may suppose from now on that $x \in \mathbb{N}$.

Let $C > 0$ be an admissible constant in Lemma 4, and choose an integer $Y \geq 2$ such that

$$\log Y > 2C.$$

For $m \geq 1$, put

$$R_Y(m) := \log m \sum_{q \geq Y} \frac{\Lambda(q)}{q \log^2(mq)}.$$

By Lemma 4 with $y = Y$, for $m \geq 2$,

$$R_Y(m) = \frac{\log m}{\log(mY)} + O\left(\frac{1}{\log m}\right) = 1 - \frac{\log Y}{\log(mY)} + O\left(\frac{1}{\log m}\right).$$

If $m \geq Y$, then $\log(mY) \leq 2 \log m$, so for all sufficiently large m ,

$$R_Y(m) \leq 1 - \frac{\log Y}{2 \log m} + \frac{C}{\log m} \leq 1.$$

Fix x_0 so large that this holds for every $m \geq x_0$, and let $x \geq x_0$.

For $m \geq x$ and integers $q \geq Y$, define

$$p(m, mq) := \frac{\log m}{\log^2(mq)} \frac{\Lambda(q)}{q}.$$

Then

$$\sum_{q \geq Y} p(m, mq) \leq 1 \quad (m \geq x).$$

Hence these numbers define a sub-Markov chain on $\{n \in \mathbb{N} : n \geq x\}$: from m the chain jumps to mq with probability $p(m, mq)$, and with the remaining probability it jumps to a cemetery state and stops. Every jump is strict, and every visited state divides every later state.

Next define the *source mass*

$$b_x(n) := \frac{1}{n \log^2 n} \left(\sum_{\substack{q|n \\ q < Y}} \Lambda(q) + \sum_{\substack{q|n \\ q \geq Y \\ n/q < x}} \Lambda(q) \right) \quad (n \geq x),$$

and let

$$B_x := \sum_{n \geq x} b_x(n).$$

We first show that

$$(2) \quad B_x = 1 + O\left(\frac{1}{\log x}\right).$$

Write $B_x = B_x^{\text{small}} + B_x^{\text{entry}}$, corresponding to the two sums in the definition of $b_x(n)$.

For the small-prime-power part,

$$B_x^{\text{small}} = \sum_{q < Y} \frac{\Lambda(q)}{q} \sum_{m \geq x/q} \frac{1}{m \log^2(mq)}.$$

Since $t \mapsto 1/(t \log^2(tq))$ is decreasing,

$$\sum_{m \geq x/q} \frac{1}{m \log^2(mq)} \leq \frac{1}{(x/q) \log^2 x} + \int_{x/q}^\infty \frac{dt}{t \log^2(tq)} \ll \frac{1}{\log x}.$$

As Y is fixed,

$$B_x^{\text{small}} = O\left(\frac{1}{\log x}\right).$$

For the first-entry part,

$$B_x^{\text{entry}} = \sum_{m < x} \frac{1}{m} \sum_{q \geq \max(Y, x/m)} \frac{\Lambda(q)}{q \log^2(mq)}.$$

For $1 \leq m < x$, set

$$y_m := \max(Y, x/m).$$

Then $my_m = \max(mY, x) \in [x, xY]$, so Lemma 4 gives

$$\sum_{q \geq y_m} \frac{\Lambda(q)}{q \log^2(mq)} = \frac{1}{\log(my_m)} + O\left(\frac{1}{\log^2 x}\right) = \frac{1}{\log x} + O\left(\frac{1}{\log^2 x}\right),$$

uniformly in m . Hence

$$B_x^{\text{entry}} = \sum_{m < x} \frac{1}{m} \left(\frac{1}{\log x} + O\left(\frac{1}{\log^2 x}\right) \right) = \left(\frac{1}{\log x} + O\left(\frac{1}{\log^2 x}\right) \right) \sum_{m < x} \frac{1}{m}.$$

Since $\sum_{m < x} 1/m = \log x + O(1)$, we obtain

$$B_x^{\text{entry}} = 1 + O\left(\frac{1}{\log x}\right).$$

Together with the estimate for B_x^{small} , this proves (2).

Now normalize the source mass by setting

$$\mu_x(n) := \frac{b_x(n)}{B_x} \quad (n \geq x).$$

Start the above sub-Markov chain with initial distribution μ_x , and for $n \geq x$ let $v_x(n)$ be the probability that the chain ever visits n .

Because the chain is strictly increasing, to visit n it must either start at n , or else arrive there in one last jump from some parent n/q with $q \geq Y$. Therefore

$$(3) \quad v_x(n) = \mu_x(n) + \sum_{\substack{q|n \\ q \geq Y \\ n/q \geq x}} v_x(n/q) p(n/q, n).$$

We claim that

$$(4) \quad v_x(n) = \frac{1}{B_x n \log n} \quad (n \geq x).$$

This is proved by induction on n . Assume (4) holds for all smaller integers. Then by (3),

$$\begin{aligned} v_x(n) &= \frac{1}{B_x} \left(b_x(n) + \sum_{\substack{q|n \\ q \geq Y \\ n/q \geq x}} \frac{1}{(n/q) \log(n/q)} \cdot \frac{\log(n/q)}{\log^2 n} \frac{\Lambda(q)}{q} \right) \\ &= \frac{1}{B_x n \log^2 n} \left(\sum_{\substack{q|n \\ q < Y}} \Lambda(q) + \sum_{\substack{q|n \\ q \geq Y \\ n/q < x}} \Lambda(q) + \sum_{\substack{q|n \\ q \geq Y \\ n/q \geq x}} \Lambda(q) \right) \\ &= \frac{1}{B_x n \log^2 n} \sum_{q|n} \Lambda(q) = \frac{1}{B_x n \log n}, \end{aligned}$$

by (1). This proves (4).

Finally, let $A \subset \mathbb{N} \cap [x, \infty)$ be primitive. Along any sample path, the visited states form a divisibility chain

$$n_0 \mid n_1 \mid n_2 \mid \cdots,$$

so the path can meet A at most once. Hence

$$\sum_{a \in A} v_x(a) \leq 1.$$

Using (4) and then (2), we conclude that

$$\sum_{a \in A} \frac{1}{a \log a} = B_x \sum_{a \in A} v_x(a) \leq B_x = 1 + O\left(\frac{1}{\log x}\right).$$

This proves the theorem. □